

Cell Type Bacterial, gram negative

Molecules Electroporated DNA: pUC 18

Species Used *E. coli*, DH10B

Before the Pulse

Cell Growth Medium 2x YT broth

Growth Phase at Harvest O.D. (600) = 0.8

Pre-pulse Incubation 0° C

Wash Solution Distilled water / 10% glycerol

The Pulse

Instruments Used Gene Pulser® apparatus and Pulse Controller

Electroporation Temperature 0 °C

Electroporation Medium 10% glycerol

Cuvette Gap 0.1 cm

Cell Density 1 x 10 (9) cells / ml

Voltage 2.5 kV

Volume of Cells 50 µl

Field Strength 25 kV/cm

DNA Concentration TE buffer

Capacitor 25 µF

DNA Resuspension Buffer TE buffer

Resistor 200 Ω (Pulse Controller)

Volume of DNA 1 µl

Time Constant 4 to 5 msec

After the Pulse

Outgrowth Medium SOC

Relevant Publications and/or Comments

Note: exponential values designated in parentheses.
SOC: 2% Bacto tryptone, 0.5% Bacto yeast extract, 10mM NaCl, 2.5mM KCl, 10 mM MgCl₂, 10 mM MgSO₄, 20 mM glucose.
2xYT: 1.6% Bacto tryptone, 1.0% Bacto yeast extract, 0.5% NaCl.

Outgrowth Temperature 37 °C

Length of Incubation 1 hour

Selection Method or Assay Used Ampicillin resistance

Electroporation Efficiency 5 x 10 (9) transformants / µg DNA

Per Cent Survival not given

Name of Submitter Motoyasn Odera

Institution Address Lion Corporation
 Biological Science Lab
 202 Tajima
 Odawara City
 JAPAN

Survey Number
 027